Yuhe Wu(吴宇赫)



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Education

Hong Kong University of Science and Technology (Guangzhou), Ph.D. in Fintech — Sep 2026 – Present
Dongbei University of Finance and Economics, Bachelor in Economic Statistics — Sep 2022 – Jun 2026

A Brief Introduction

I am a **Ph.D. student** in Fintech at the **Hong Kong University of Science and Technology (Guangzhou)**, supervised by Prof. Guang Zhang. I received my B.S. in Economic Statistics from Dongbei University of Finance and Economics (DUFE), where I was supervised by Prof. Zhuang Liu and Prof. Qiang Xu. My research focuses on **LLM capabilities**, spanning **LLM evaluation**, **LLM post-training**, and **generative agents**. I have published papers at venues including **ACL 2026**, **EMNLP 2026**, **KDD 2025**, and **Annals of Operations Research**, with additional manuscripts under review at **WWW 2027**, **AAAI 2027**, **ACL 2027**, and **Management Science**. I also serve as a reviewer for leading venues such as **ACL**, **KDD**, **NeurIPS**, **AAAI**, and **IJOC**.

Research Publications[1]

LLM Evaluation

This line evaluates and interprets large language models beyond traditional benchmarks, asking where in generation factuality and reasoning fail, why they fail, and how those insights can guide the next training step. PRISM probes hallucinations across memory, instruction, and reasoning. BizCompass measures whether disciplinary knowledge becomes performance in a business role. Caught in the Story observes failures in multi-turn conversation and attributes them to a specific stage of post-training. MIND benchmarks long-form value measurement across public events and value dimensions. The graph question-answering benchmark extends this evaluation to open-set questions over text attribute time graphs.

- **Yuhe Wu**, Guang Zhang, Zhuang Liu, et al. *PRISM: Probing Reasoning, Instruction, and Source Memory in LLM Hallucinations.* In **ACL 2026 Main Conference** *(CCF-A, CORE A*).*
- **Yuhe Wu**, Guang Zhang, Zhuang Liu, et al. *Caught in the Story: Narrative Captivity in Multi-turn LLMs Conversation.* In **EMNLP 2026 Findings** *(CCF-B, CORE A).*
- Jianing Hao*, **Yuhe Wu***, Yuanjian Xu*, Guang Zhang, et al. *BizCompass: Benchmarking the Reasoning Capabilities of LLMs in Business Knowledge and Applications.* In **ACL 2026 Findings** *(CCF-A, CORE A*; *Equal contribution).*
- Guangyu Wang, **Yuhe Wu**, Jiawei Wang. *Benchmarking open set question answering in text attribute time graphs.* **Under review at WWW 2027** *(CCF-A, CORE A*).*

LLM Post-Training

Post-training here centers on adaptation of language models—making them precisely steerable, and more capable without making them less reliable. Supervision in this setting is always imperfect; preference and reward signals shape behavior, and self-improving post-training is how that steering is kept. COR^2 shapes the reward around the structure of an optimization model rather than its outcome. PackRank makes concise reasoning the default by selecting short, accurate traces from packed queries for self-training. BizSage turns self-improvement toward automated business research. Don't Guess the User keeps ambiguity explicit through set-valued semantics and value-of-information clarification, while Vis-MCTS uses LLM-guided search for training-free algorithm discovery.

- **Yuhe Wu**, Guang Zhang, et al. *BizSage: A Self-Evolving Multi-Agent Framework for Business Research with Efficient Knowledge Retrieval.* In **EMNLP 2026 Findings** *(CCF-B, CORE A).*
- **Yuhe Wu**, Zhuang Liu, et al. *Enhancing Financial Decision-Making under Cyber Threats: A Dual-Branch Framework Integrating Bayesian Deep Learning and Explainable AI.* In ***Annals of Operations Research*** *(ABS 3, JCR Q1).*
- **Yuhe Wu**, Guang Zhang, et al. *Spend Reasoning Where It Counts: Concise LLM Operations Research Modeling.* **Under review at AAAI 2027** *(CCF-A, CORE A*).*

[1] • Published; • Under review; • Ongoing

- **Yuhe Wu**, Rui Qian, Yanzhen Chen, Guang Zhang. *PackRank: Learning Concise Reasoning Rerankers from Packed Queries.* **Under review at ACL 2027** *(CCF-A, CORE A*).*
- Wei Wu, Rui Qian, **Yuhe Wu**†. *Vis-MCTS: LLM-Driven Autonomous Discovery of Training-Free Wide-Angle Correction Algorithms.* **Under review at ACL 2027** *(CCF-A, CORE A*; †Corresponding author).*
- Haoyu Wang, **Yuhe Wu**, Jiachen Luo, Hao Tang. *Don't Guess the User: Set-Valued Semantics and Value-of-Information Clarification for Tool-Using LLM Agents.* **Under review at ACL 2027** *(CCF-A, CORE A*).*

Generative Agents

The work studies how LLM-based agents perceive, reason, interact, and adapt in multi-agent environments, and how such simulations can be tested against how people perceive, decide, and act. SEDM aligns how agents make decisions in social interaction, rather than the surface answer. CasT^2 simulates collective behavior from the content that starts it. DND combines node-level LLM agents with graph-based action selection for fair resource allocation in large-scale dynamic networks. Headwater diagnoses who failed, by which pathology, and from which round. Where Conditions Get Lost tests whether generated social networks preserve conditional structure across people and contexts. The affective attachment framework models how memory, empathy, and accumulated affect shape loyalty and switching behavior in academic hierarchies.

- Jiaxin Liu*, **Yuhe Wu***, Guang Zhang, et al. *Mechanism Alignment for Socially Embedded Decision Making in LLMs.* In **EMNLP 2026 Main Conference** *(CCF-B, CORE A; *Equal contribution).*
- **Yuhe Wu**, Guangyu Wang, Guang Zhang. *What's in the Water: Auditing Faults in LLM Social Simulations.* **Under review at ACL 2027** *(CCF-A, CORE A*).*
- Jiaxin Liu*, **Yuhe Wu***, Yu Wang, Yixuan Xie, Siguang Li, Guang Zhang. *Where Conditions Get Lost: Conditional Validity of LLM-Generated Ego Networks.* **Under review at ACL 2027** *(CCF-A, CORE A*; *Equal contribution).*
- Guangyu Wang*, **Yuhe Wu***, Zhaonan Wang. *Beyond When: How Root Content Propagates as Information Cascade Trees.* **Under review at WWW 2027** *(CCF-A, CORE A*; *Equal contribution).*
- Guangyu Wang*, **Yuhe Wu***, Jiawei Wang. *DND: A Decision Engine for Large-Scale Dynamic Networks.* **Under review at WWW 2027** *(CCF-A, CORE A*; *Equal contribution).*
- Jia Yee Tan, Yusen Wu, **Yuhe Wu**†. *A Generative Agent Framework for Simulating Affective Attachment in Academic Hierarchies.* **Under review at ACL 2027** *(CCF-A, CORE A*; †Corresponding author).*

Research Experience

Fintech Thrust, HKUST(GZ), Research Assistant, Supervised by Prof. Guang Zhang — Jun 2025 – Aug 2026

National Accounting Center, DUFE, Group Core Member, Supervised by Prof. Qiang Xu — Jan 2025 – Jul 2026

Fintech Lab, DUFE, Group Leader, Supervised by Prof. Zhuang Liu — Mar 2024 – Jul 2026

Honors & Awards

- ACL 2026 Diversity & Inclusion Grant (**TOP 1%**) — 2026
- KDD-25 Undergraduate Scholarship (**TOP 1%**) — 2025
- Dashang Group Academic Research Scholarship (**TOP 1%**) — 2025
- Excellent Innovation and Entrepreneurship Team Scholarship (**TOP 1%**) — 2025
- First-Class Scholarship (**TOP 2%**) — 2023 & 2024
- Core Member, Liaoning Marine Economy Study (5th National Economic Census Project) — 2024
- Core Member, Text–Price Correlation in Food Industry (Liaoning Edu. Dept. Innovation Grant) — 2024

Skills

Machine Learning: LLM evaluation and post-training, including supervised fine-tuning, RLHF, and preference optimization; generative models and agent frameworks; hands-on experience with DeepSeek-R1, Qwen, and Llama.

Statistical Analysis: Statistical analysis and modeling with SPSS, Stata, SAS, and R.

Research Tools: LaTeX, Linux, Conda, Git, Origin, and Visio.